Sadguru Balumama Shikshan Prasarak Mandal's

K.P.PATIL INSTITUTE OF TECHNOLOGY

(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai &Govt. of Maharashtra,Affiliated to MSBTE Mumbai

Internships Daily Diary(2025-26)

Department Of Artificial Intelligence & Machine Learning Engineering

Program Code :AI5K

Name of the Student : Saniko Dhanaji Chougale

Name of the Mentor (Faculty) : Mrs. Poonam Kumbhar

Name of the Mentor (Industry): Mr. Abhijit Gatade

Enrollment Number : 23213500044

Week	Day & Date	Discussion Topics/Activity	Details of Work Allotted Till Next Session /Corrections Suggested/Faculty Remarks	Signature of Industry Mentor
Week 10	Mon, Date 04/08/25	Understanding project and dataset features.	Understood project aim and dataset.	
	Tue, Date 05/08/25	Data cleaning and preprocessing step.	Cleaned and prepatterns & corrections	
	Wed, Date 06/08/25	~~B~~ Performing exploratory data analysis	Explored data patterns and corrections.	
	Thu, Date 07/08/25	Building logistic analysis regression model	Trained and evaluated logistic regression	
	Fri, Date 08/08/25	Implementing random forest classifier	Built and turned random forest model	
	Sat, Date 9/08/25	Saving final model and reflections.	Saved model and documented workflow.	

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Internships Daily Diary (2025-26)

Week No: 10 **Day:** Monday **Date:** 04/08/25

- Understanding Project and Dataset.

The project on crop yeild prediction begins with understanding its importance in agriculture and food security. crop yeild refers to the amount of crop produced per unit area and is influenced by multiple environmental and agricultural factors such as rainfall, temperature, Soil nutrients, fertillizer usage, and crop type. In this project, the dataset includes both numerical both numerical features (e.g. rainfall, NPK values, temp). and categorical features. The target variable is crop yeild, usually measured in kilograms per hecature. Theoretical understanding of the dataset is crucial because machine learning models rely on patterns.

Internships Daily Diary (2025-26)

Week No: 10 **Day:** Tuesday **Date:** 05/08/25

- Data Cleaning and preprocessing

Data cleaning is critical step because real-world agricultural datasets are often incomplete, inconsistent, or noisy. Missing values in rainfall or soil nutrients need to be addressed using impudtation techniques such as mean, median, or interpolation. Outliers in fertilizer use or yeild values must be identified because they can skew predictions. Categorical variables like crop type or soil type are transformed into numerical from, through encoding methods such as One-Hot Encoding Numerical features like rainfall and fertilizer are normalized or scaled.

Internships Daily Diary (2025-26)

Week No: 10 **Day:** Wednesday **Date:** 06/08/25

- Exploratory Data Analysis

Exploratory Data Analysis is the process of examining the dataset through visualization and stustical method to uncover hidden patterns. It allow researchers to understand how different features influence crop yeild. Scatterplots can be used to visudlize relationships such as rainfall versus yeild. Histograms show the distribution of Soil nutrient levels, while boxplots hightlight Variability and Outliers in Crop production across regions. A Correclation matrix identifies the strength of relationships among Variables, which is crucial for Selecting features that significantly affect yeild.

Internships Daily Diary (2025-26)

Week No: 10 **Day:** Thusday **Date:** 07/08/25

- Building Linear Regression Model.

Linear Regression is the most basic Sapervised learning algorithm used to predict Continuous variable such as crop yeild. It establishes a linear relationship between independent variables and the dependent variables.

The mathematical equation takes the form $Y = ax + b$, where coeffci-ents. represent the influence of each factor on yeild. During training, the model fits the best possible line that minimize the error between actual and predicted values, performance is evaluated using metrics such as the R^2 score, Mean Absolute Error (MAE), Root Mean Square error.

Internships Daily Diary (2025-26)

Week No: 10 **Day:** friday **Date:** 08/08/25

- Implementing Random Forest Model.

Random forest is an advanced ensemble learning technique that overcomes the limitations of Linear Regression by handling nonlinear and Complex interactions between Variables. It works by constructing multiple decision trees during training and combining their result to improve accuracy and reduce overfitting. each tree considers different Subsets of Features, ensuring robustness in prediction. Hyperparameters such as the number of trees, maximum depth, and minimum samples per split are tuned to achieve the best performance.

Internships Daily Diary (2025-26)

Week No: 10 **Day:** Saturday **Date:** 09/08/25

- Saving final Model and Reflections:

Once the best-performing model is developed, it is saved using tools like joblib or pickle for future use and deployment. Saving the model ensures that it can be integrated into applications for real-time predication without retraining. A complete documentation of the workflow is prepared, covering dataset preprossing, exploratory analysis, model building, evaluation and result. finally, the completed project lays the foundation for further improvements using neural networks or deployment in web-based platforms.